

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 9: Human Error & Reliability Models

Syllabus Source: CAE Oxford (pp. 279-318) | Pass Target: >= 90% AviationExam

1. HUMAN ERROR TAXONOMIES: RASMUSSEN'S SRK MODEL

Jens Rasmussen classified human cognitive behavior and associated error mechanisms into three distinct levels:

Performance Level	Cognitive Nature & Cockpit Context	Characteristic Error Mechanisms
Skill-Based (SB)	Automated sensory-motor performance (flying manual approach, checklist scanning).	SLIPS (action execution failure - unintended switch selection) and LAPSES (memory failure - skipped checklist item due to distraction). Occurs in familiar, practiced routines.
Rule-Based (RB)	Applying established IF-THEN procedures, rules, or checklists to a known problem.	RULE-BASED MISTAKES: Selecting the wrong rule, or correctly applying a bad/flawed rule. High confidence, but incorrect diagnosis of situation.
Knowledge-Based (KB)	Novel, unfamiliar emergency scenarios with no pre-existing rule or SOP (unprecedented crisis).	KNOWLEDGE-BASED MISTAKES: Incomplete or incorrect mental model of the system. Highest mental workload and cognitive effort. High probability of error.

2. ERRORS VS VIOLATIONS (INTENTIONALITY SPECTRUM)

- Human Error: Unintentional deviation. The planned action fails to achieve the desired outcome (Slips, Lapses, Mistakes). Non-punitive under Just Culture.
- Routine Violation: Deliberate shortcutting of SOPs commonly condoned by peer group or company culture (e.g. not calling for official checklist, speeding on taxiways).
- Situational Violation: Deliberate rule infringement driven by commercial pressure, poor equipment, or time deficit ('getting the job done').
- Exceptional Violation: One-off isolated violation occurring in an unprecedented emergency to save life or aircraft.

3. JAMES REASON'S SWISS CHEESE MODEL (DEFENSES-IN-DEPTH)

Catastrophes in complex sociotechnical systems never result from a single isolated human error. They occur when defensive barriers across multiple organizational and operational layers breach simultaneously:

Barrier / Cheese Layer	Failure Nature	Aeronautical Manifestation & Trigger Mechanism
Organizational Influences	Latent Conditions	Corporate cost-cutting, inadequate training budget, flawed rostering, unrealistic flight schedules.
Unsafe Supervision	Latent Conditions	Tolerating known SOP non-compliance, inadequate flight crew pairing, failure to address maintenance reports.
Preconditions for Unsafe Acts	Latent / Hybrid	Crew fatigue, high workload, emotional stress, poor cockpit ergonomic layout, confusing checklists.

Unsafe Acts (Active Failures)**Active Failures**

Immediate operational errors committed by front-line personnel (pilots, ATCOs): pilot flies below MDA, misses readback error.

[!] EXAM TRAP (AVIATIONEXAM) - Latent Conditions vs Active Failures (AviationExam Absolute Core)

- Latent Conditions: Decisions made by management, regulators, designers, or maintainers that lie DORMANT in the system for months or years without causing harm, until triggered by operational circumstances. - Active Failures: Unsafe acts committed by operational personnel at the sharp end (pilots, controllers) whose consequences are felt ALMOST IMMEDIATELY.

4. THE SHELL MODEL & THREAT AND ERROR MANAGEMENT (TEM)

- SHELL Model (Hawkins / Edwards): Software (rules, SOPs, manuals), Hardware (airframe, cockpit displays), Environment (weather, night, noise), Liveware (the human operator at the center, interacting with Liveware-Others - crew, ATC). - Threat and Error Management (TEM): A proactive operational framework dividing operations into Threats (environmental/technical), Errors (crew deviations), and Undesired Aircraft States (UAS - deviations requiring immediate recovery before an accident occurs).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis**

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: According to Jens Rasmussen's SRK model, a pilot who inadvertently operates the landing gear switch in

- [A]: A rule-based mistake
- [B]: A knowledge-based mistake
- [C]: A skill-based slip of action
- [D]: An exceptional violation

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CORRECT: [C]. Operating the wrong control in a familiar automated motor routine is a classic slip of action (skill-based level).

Q2: In James Reason's Swiss Cheese model, what distinguishes a 'latent condition' from an 'active failure'?

- [A]: Latent conditions always cause instant crashes
- [B]: Active failures are organizational; latent conditions are individual
- [C]: Latent conditions are dormant systemic defects created by management or design, whereas active failures are immediate unsafe acts committed by front-line operators
- [D]: Latent conditions only apply to maintenance

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CORRECT: [C]. Latent conditions exist dormant within the system (rostering, training, SOP design), triggered into an accident by an active front-line operator failure.

Q3: In the SHELL model of human factors, what is positioned at the central core of the system?

- [A]: Hardware (the aircraft avionics)
- [B]: Software (the operating procedures and regulations)
- [C]: Liveware (the human being)
- [D]: Environment (the atmospheric weather)

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CORRECT: [C]. Liveware (the human operator) sits at the center, interacting with Software, Hardware, Environment, and other Liveware.

Q4: A flight crew deliberately decides to omit a mandatory before-landing checklist to expedite approach due to time pressure.

- [A]: An involuntary lapse
- [B]: A situational / routine violation
- [C]: A skill-based slip
- [D]: A knowledge-based error

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CORRECT: [B]. A conscious, intentional departure from published rules under time or operational pressure is classified as a violation.

Q5: Under Threat and Error Management (TEM), what is an 'Undesired Aircraft State' (UAS)?

- [A]: An airline undergoing bankruptcy
- [B]: An aircraft configuration or position deviation caused by an unmanaged error, which reduces safety margins
- [C]: A bird strike during take-off
- [D]: Moderate clear air turbulence

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CORRECT: [B]. A UAS is an operational deviation (e.g. unstable approach, wrong altitude) that must be recovered immediately to prevent an accident.

Q6: At which cognitive performance level does human conscious mental workload reach its maximum and have the greatest impact on performance?

- [A]: Skill-based level
- [B]: Rule-based level
- [C]: Knowledge-based level
- [D]: Automated reflex level

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CORRECT: [C]. The knowledge-based level requires deep analytical problem solving in novel emergency scenarios, placing extreme demands on working memory.